

Virtual Memory In Class Assignment

1. List 2 advantages of using a virtual address space as opposed to the physical address space.

We can write programs that are bigger than our physical memory space, and we can also run more programs since each program will take less physical space in memory.

2. What is meant by the term 'lazy swapper'? List some advantages of using this technique for memory management.

A lazy swapper will never swap a page into memory unless that page is needed. This is beneficial because we can keep more frames open as pages that are never accessed, are never loaded into the system. Additionally, since only the pages that are needed are loaded, there's more physical space free since we aren't loading all the pages in order to run the process. Plus, there's less overhead since we don't have to move a huge amount of pages since some pages may never be needed.

3. What is the valid/invalid bit used for in the page table?

It's meant to tell us where a page is within our machine. A valid bit means that the page is in physical memory, an invalid bit means the page is not in physical memory.

4. What is meant by the term 'copy-on-write'? How is this approach advantageous to memory management?

Copy on write is referring to how a page can be initially shared by two processes. However, the page is only copied, once it needs to be written. This is advantageous because if we fork and both processes still reference the original page in memory, we can save space by just having them continue to reference the same page. Once one makes changes to the page, then we make a copy, this way if changes are never made, then we save space. Plus, until those changes are made, we save more space than we would've if we always immediately copy.

5. Given the reference string used in the class lecture and textbook, compare the performance (page faults) of each of the following page replacement algorithms assuming the process has 3 pages available to it:

1. FIFO

701
201
201
231
230
430
420
423
023
023
023
013
012
012
012
712
702
701

2. LRU

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

X, x, x, x, x, x, x, x, x, x, x, x, x, x, x, x, x, x x, x

701

201

201

203

203

403

402

432

032

032

032

132

132

102

102

107

107

107

3. Second Chance

1

$$7_0 0_0 1_0$$
$$2_0 0_0 1_0$$
$$2_0 0_1 1_0$$

2030

$$2_0 0_1 3_0$$

4₀0₀3₀

$4_0 2_0 3_1$
 $4_0 2_0 3_0$
 $0_0 2_0 3_0$
 $0_0 2_1 3_1$
 $1_0 2_0 3_0$
 $1_0 2_1 3_0$
 $1_0 2_1 0_0$
 $1_1 2_1 0_0$
 $1_0 2_0 7_0$
 $1_0 0_0 7_0$
 $1_1 0_0 7_0$

4. Optimal Page Replacement

Reference String:

0, 1

701
201
201
203
203
243
243
243
203
203
203
201
201
201
201
701
701
701

6. (optional) If time permits in class, create your own algorithm for page replacement. Describe it below and check it's performance using the above reference string (if able).